Euler's Theorem: Comprehensive Explanation & Solutions

1. Initial Problem Solution (Question 15)

Problem Statement: If $u = \text{cosec}^{-1} [(x^{1/2} + y^{1/2}) / (x^{1/3} + y^{1/3})]^{1/2}$, prove that:
$x^2(u_{xx}) + 2xy(u_{xy}) + y^2(u_{yy}) = [\tan(u) / 12] * [13/12 + \tan^2(u)/12]$

Step 1: Define a homogeneous function
Rearranging and squaring both sides:
$z = \text{cosec}^2(u) = (x^{1/2} + y^{1/2}) / (x^{1/3} + y^{1/3})$

Step 2: Determine the degree of homogeneity (n)
The degree of the numerator is 1/2, and the denominator is 1/3.
$n = 1/2 - 1/3 = 1/6$

Step 3: Apply First Order Euler's Theorem
By Euler's Theorem: $x(z_x) + y(z_y) = nz$
Differentiating $z = \text{cosec}^2(u)$ implicitly via Chain Rule:
$z_x = -2 \text{cosec}^2(u) \cot(u) u_x$
Substituting these back yields standard first-order relation:
$x(u_x) + y(u_y) = -1/12 * \tan(u) = g(u)$

Step 4: Use the Second-Order Deduction
The second-order form states: $x^2(u_{xx}) + 2xy(u_{xy}) + y^2(u_{yy}) = g(u)[g'(u) - 1]$
Differentiating $g(u) = -1/12 * \tan(u)$ gives $g'(u) = -1/12 * \sec^2(u)$.
Substituting both values and using identity $\sec^2(u) = 1 + \tan^2(u)$ simplifies to:
$[\tan(u)/12] * [13/12 + \tan^2(u)/12]$ (Hence Proved).

2. Conceptual Breakdown: Different Degrees in Numerator & Denominator

A function is homogeneous if scaling all independent variables by a constant factor *t* scales the entire function uniformly by t^n.

Let $z(x, y) = (x^{1/2} + y^{1/2}) / (x^{1/3} + y^{1/3})$
Substitute $x \to tx$ and $y \to ty$:
$z(tx, ty) = [(tx)^{1/2} + (ty)^{1/2}] / [(tx)^{1/3} + (ty)^{1/3}]$
Factoring out *t* terms:
$z(tx, ty) = [t^{1/2} (x^{1/2} + y^{1/2})] / [t^{1/3} (x^{1/3} + y^{1/3})]$
Using exponent rules ($t^a / t^b = t^{a-b}$):
$z(tx, ty) = t^{(1/2 - 1/3)} * z(x, y) = t^{1/6} * z(x, y)$

Since the function scales uniformly, Euler's theorem remains legally applicable with a fractional degree n = 1/6.

3. Implicit Differentiation & The Chain Rule

Why can't we treat $\text{cosec}^2(u)$ as a constant relative to x? Because **u is a dependent function of both x and y** (u = u(x,y)).

Simpler Illustrative Example:
Let $u = x^2 + y^2$ and $z = u^3$.
If we use direct substitution:
$z = (x^2 + y^2)^3 \Rightarrow \partial z/\partial x = 3(x^2 + y^2)^2 * 2x = 6x(x^2 + y^2)^2$
If we use implicit differentiation (Chain Rule):
$\partial z/\partial x = (dz/du) * (\partial u/\partial x) = (3u^2) * (2x) = 6xu^2$
Substituting u back gives the same answer: $6x(x^2 + y^2)^2$.
Treating u as a constant would mistakenly yield 0, missing the real rate of change.

4. Second-Order Equations of Euler's Theorem

Case 1: Standard Algebraic Function z(x, y)
If z is a homogeneous function of degree n:
$x^2(\partial^2 z/\partial x^2) + 2xy(\partial^2 z/\partial x\partial y) + y^2(\partial^2 z/\partial y^2) = n(n - 1)z$

Case 2: Composite Function u(x, y) where f(u) = z
If you have established the first-order relation $x(\partial u/\partial x) + y(\partial u/\partial y) = g(u)$, then:
$x^2(\partial^2 u/\partial x^2) + 2xy(\partial^2 u/\partial x\partial y) + y^2(\partial^2 u/\partial y^2) = g(u)[g'(u) - 1]$